



TECHNICAL BRIEF: TARGETING AGRICULTURAL LIME FOR ACID CROPLANDS IN ETHIOPIA

GAIA

Last updated: November 18, 2024



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INTRODUCTION

This technical brief provides a summary of the data-driven, ex-ante analysis of soil acidity in ETHIOPIA, assessing the extent of affected areas, potential impacts on crop yields, and the economic benefits of lime application for soil remediation. Soil acidity poses a significant threat to agricultural productivity, particularly in regions with widespread acidic croplands. The findings presented here aim to offer policymakers, farmers, and stakeholders actionable insights into prioritizing areas for lime application to maximize productivity gains and economic returns.

Significant portions of ETHIOPIA's cropland are affected by soil acidity-related constraints. Recent investments in national and regional soil information systems, such as [Country-specific Soil Information System], have generated updated spatial estimates of soil properties, making soil acidity a priority topic for agricultural development in the country. In response, discussions have emerged around the potential mobilization of investment resources to initiate large-scale liming campaigns in regions affected by acid soils. To support and guide these efforts, research initiatives involving [Country-specific Research Institute] and the International Maize and Wheat Improvement Center (CIMMYT), under the Guiding Acid Soils Management Investments in Africa (GAIA) project supported by the Bill and Melinda Gates Foundation, have focused on soil acidity remediation through agricultural lime and other practices.



FIGURE 1: SPATIAL DISTRIBUTION OF SOIL ACIDITY IN ETHIOPIA

The purpose of this technical brief is to (a) summarize key research findings on acid soil management and crop yield response to lime and (b) guide soil acidity-related investments and strategic policy in ETHIOPIA. The brief draws on analysis of lime trials data collected from different parts of the country over the last two years, as well as broader-scale spatial analysis designed to inform targeting of remediation efforts at scale across ETHIOPIA.

HIGHLIGHTS

- **Acidic Croplands:** Identification of regions with high soil acidity, negatively impacting agricultural output. In ETHIOPIA, [X]% of the total cropland is classified as acidic, requiring immediate intervention.

- ▶ **Yield Loss:** Estimated yield losses for key crops such as maize, sorghum, and wheat due to soil acidity. On average, acidic soils in ETHIOPIA result in yield reductions of approximately [Y]%, affecting food security and economic stability.
- ▶ **Lime Requirements:** Estimates of the quantity of lime required to rehabilitate acidic soils and achieve optimal crop yields. Preliminary estimates indicate a need for [Z] thousand tonnes of lime in the most affected regions.
- ▶ **Economic Potential:** Projected increase in crop production and production value with lime application. Application of lime is expected to increase annual crop yields by [A]% and generate an additional economic value of [B] million USD.



FIGURE 2: YIELD RESPONSE TO LIME APPLICATION IN ETHIOPIA

SOIL ACIDITY AND ITS IMPACTS ON AGRICULTURE

Soil acidity significantly reduces the productivity of agricultural land by limiting nutrient availability and impairing root growth. In ETHIOPIA, soil acidity is a major constraint on crop production, particularly in key agricultural regions. Acidic soils result in poor crop establishment, reduced nutrient uptake, and ultimately lower yields. Addressing soil acidity through lime application can help alleviate these constraints and improve crop health, leading to more resilient agricultural systems.

Recent analysis of soils in ETHIOPIA shows that many areas with low pH levels are characterized by high exchangeable acidity, which impedes root growth and nutrient uptake by crops. Exchangeable acidity can be most effectively addressed through lime application. However, in regions where acidity is low, other approaches such as integrated soil fertility management (ISFM) that combine mineral and organic fertilizer inputs may be more cost-effective.

SPATIAL DISTRIBUTION OF SOIL ACIDITY

The distribution of exchangeable acidity across ETHIOPIA is heterogeneous, with hotspots of high acidity levels predominantly found in [Region A], [Region B], and

[Region C]. Exchangeable acidity, expressed as a percentage of the Effective Cation Exchange Capacity (ECEC), indicates the need for targeted lime interventions to restore soil productivity in these areas. Mapping the spatial distribution of soil acidity helps identify priority areas for lime application, allowing for efficient allocation of resources.

Recent mapping suggests that areas with pH below 5.5 are candidates for liming campaigns in ETHIOPIA, as low pH is often associated with high exchangeable acidity. However, some areas with low pH may not have high exchangeable acidity, indicating that soil fertility management may be a better solution. Prioritizing regions with exchangeable acidity above 10% of ECEC is recommended, as these areas are more likely to experience severe yield losses due to acidity.

LIME REQUIREMENTS AND ECONOMIC BENEFITS

The estimated lime requirement to remediate acidic soils in ETHIOPIA is approximately [Z] thousand tonnes. This estimation was derived using the Aramburu Merlos et al. (2022) lime requirement model, applied to gridded soil data from Hengl et al. (2017). The cost of lime application is calculated based on a fixed lime price of 100 USD per tonne, while the economic return is computed using median crop prices from sub-Saharan Africa.



FIGURE 3: LIME REQUIREMENTS FOR ACID SOIL REMEDIATION IN ETHIOPIA

The benefits of lime application are substantial, particularly in areas such as [Region X] and [Region Y], where the increase in yield potential translates into economic gains of [B] million USD. The positive response of crop yields to liming indicates a significant return on investment, making lime application a financially viable solution for improving agricultural productivity in acidic soils. In addition to increased yields, liming can improve soil health, enhance nutrient availability, and contribute to long-term agricultural sustainability.

Localized lime recommendations can vary substantially within regions, highlighting the importance of targeted interventions. For areas with high exchangeable acidity (>10% of ECEC), lime rates of up to 2 t/ha are recommended. For areas with low acidity, lower lime rates (less than 1 t/ha) or alternative interventions, such as ISFM, should be considered. The cost-benefit trade-offs of liming versus

supplemental phosphorus management are still under investigation, with results expected in [Year].

IMPLEMENTATION CONSIDERATIONS

- ▶ **Cost and Logistics:** The logistics of lime distribution and application are critical to the success of soil remediation efforts. In ETHIOPIA, challenges such as limited infrastructure, high transportation costs, and inadequate supply chains may hinder lime availability in remote areas. Effective planning and coordination among government agencies, private sector stakeholders, and farmer cooperatives are essential to overcome these challenges.
- ▶ **Farmer Awareness and Training:** Increasing farmer awareness of soil acidity and its impact on crop yields is crucial. Extension services and training programs should be established to educate farmers on the benefits of lime application, proper application rates, and best practices for improving soil health. Demonstration plots and farmer field schools can be used to showcase the positive effects of lime on crop performance.
- ▶ **Policy Support:** Government policies that promote lime use through subsidies, credit facilities, and incentives can help reduce the financial burden on farmers and encourage adoption. Policymakers should prioritize the development of supportive policies that facilitate lime production, distribution, and application in high-priority areas.
- ▶ **Targeted Promotion:** Recommendations for targeting lime distribution include ramping up soil testing efforts and focusing on areas with high exchangeable acidity (> 10%). Targeting specific crops, such as legumes and root and tuber crops, may also yield greater returns compared to cereals, given their different sensitivities to soil acidity.

KEY ASSUMPTIONS AND PARAMETERS

- ▶ **Acidic Cropland Definition:** Cropland with acidity saturation levels above 10% of ECEC is classified as acidic.
- ▶ **Yield Response:** Crop yields are assumed to be unaffected by soil acidity until a critical acidity threshold, known as the Target Acidity Saturation (TAS), is reached. Beyond this point, yields decrease linearly, with zero yield expected at very high acidity levels.
- ▶ **Economic Benefit Calculation:** The benefit is calculated as the additional yield from liming multiplied by the median crop price, while costs are determined by multiplying the lime rate by a fixed price of 100 USD/tonne.
- ▶ **Timeframe:** The benefits of lime application are assessed for the year of application due to limited data on the long-term effects of liming in sub-

Saharan Africa. Further research is needed to evaluate the residual benefits of lime over multiple growing seasons.

- ▶ **Crop Differences:** Different crops have varying tolerances to soil acidity, which influences lime recommendations. For cereals, 70% of cropland may not require lime, while for legumes, up to 45% of cropland could benefit from lime application.

CONCLUSION

Targeting lime application in the most affected regions of ETHIOPIA presents an opportunity to significantly increase agricultural productivity and improve food security. By addressing soil acidity, policymakers can prioritize interventions that yield the highest economic returns and ensure long-term agricultural sustainability. In addition, lime application can contribute to building resilient farming systems capable of withstanding the challenges of climate variability and soil degradation. Given the substantial logistics involved in lime distribution, it is important to focus on regions where the return on investment is highest, and consider alternative soil fertility management practices where appropriate.

For more detailed analysis and guidance, visit www.acidsoils.africa.

NEXT STEPS

- ▶ **Data Collection:** Conduct detailed soil testing in target regions to refine lime requirement estimates and ensure accurate targeting of interventions.
- ▶ **Stakeholder Engagement:** Collaborate with government agencies, NGOs, and private sector partners to develop an action plan for lime distribution and application.
- ▶ **Monitoring and Evaluation:** Establish a monitoring and evaluation framework to track the impact of lime application on crop yields, soil health, and farmer livelihoods, and adjust strategies as needed based on field observations.
- ▶ **Further Research:** Continue research into the cost-effectiveness of lime versus alternative soil fertility management practices, especially in areas with low exchangeable acidity, to ensure that recommendations are both agronomically and economically sound.